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T.E. (Computer Engg.)

WEB TECHNOLOGY

(2019 Pattern) (Semester - II) (310252)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Answer Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q7 or Q8
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right indicate full marks.
- 4) Assume suitable data if necessary.

Q1) a) Explain Http Servlet class with example. Explain the servlet Life cycle method. [10]

b) Explain XML using DTDS & XML using XML Schema with example.[8]

OR

Q2) a) Explain servlet session management & tracking methods. [9]

b) Write note on :

i) AJAX [5]

ii) do Get Vs dopost [4]

Q3) a) Explain JavaBeans classes & JSP. [8]

b) Explain the concept of Web Services. Explain WSDL & SOAP. [9]

OR

Q4) a) Explain JSP Vs Servlets (Min.04). Explain how JSPS are translated to Servlets. [8]

b) Write note on struts. Explain struts with example code block. [9]

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Q5) a) Explain the MySQL & PHP for inserting the students data into the students table. (Assume suitable data) [8]

b) Write note on :

i) ASP. NET [5]

ii) NodeJS [5]

OR

Q6) a) Explain PHP functions and arrays in PHP. [10]

b) Explain the concept of NET framework and C# [8]

Q7) a) Explain the Ruby with respect to scalar types & their operations, control statements & arrays. [12]

b) Write note on EJB. [5]

OR

Q8) a) Explain the Rails Applications & databases. Also explain Rails with AJAX. [12]

b) Explain the advantages & Ruby & Rails. [5]



Solutions:

Q1 a) Explain HttpServlet class with example. Explain servlet life cycle. (10 marks)

Ans:

1. HttpServlet Class (Introduction)

- HttpServlet is a class provided by **Java Servlet API**.
- It is used to create web applications that work using **HTTP protocol**.
- It provides built-in methods like **doGet()**, **doPost()**, **doPut()**, **doDelete()**, etc.
- Instead of writing complete request-handling code, developers only override required methods.

2. Important Methods of HttpServlet

Method	Purpose
doGet()	Handles HTTP GET requests (URLs, form with method="get")
doPost()	Handles POST requests (form with method="post")
init()	Called only once when servlet is created
service()	Calls doGet(), doPost(), etc. depending on request type
destroy()	Called when servlet is removed from memory

3. Example of HttpServlet

```
import java.io.*;
import javax.servlet.*;
import javax.servlet.http.*;

public class HelloServlet extends HttpServlet {

    public void doGet(HttpServletRequest request, HttpServletResponse response)
        throws ServletException, IOException {

        response.setContentType("text/html");
        PrintWriter out = response.getWriter();

        out.println("<h2>Welcome to HttpServlet Example!</h2>");
    }
}
```

This servlet prints a simple message when accessed through browser.

4. Servlet Life Cycle

Servlet life cycle means **how servlet is created, used, and destroyed** by the container (Tomcat).

There are **3 main stages**:

Stage 1: Initialization → **init()**

- Servlet container (Tomcat) loads servlet class for the first time.
- It creates servlet object.
- Calls **init()** method once.
- Used for:
 - Opening database connections
 - Initial configurations

Stage 2: Request Handling → **service()**

- Called every time a request comes.
- Decides which method to call:
 - GET → doGet()
 - POST → doPost()

Stage 3: Destruction → **destroy()**

- When application shuts down or servlet is no longer required:
- Container calls **destroy()**
- Used for:
 - Closing connections
 - Releasing resources

Q1 b) Explain XML using DTDs & XML using XML Schema with example. (8 marks)

Ans:

XML is used to store and transfer data.

To validate XML structure, we use **DTD** or **XML Schema**.

1. XML Using DTD (Document Type Definition)

What is DTD?

- DTD defines **structure, elements, attributes** allowed in XML.
- It checks whether XML follows correct structure.
- Written in simple rules.

Example: DTD + XML

employee.dtd

```
<!ELEMENT employee (emp_id, emp_name, emp_dept)>
<!ELEMENT emp_id (#PCDATA)>
<!ELEMENT emp_name (#PCDATA)>
<!ELEMENT emp_dept (#PCDATA)>
```

employee.xml

```
<?xml version="1.0"?>
<!DOCTYPE employee SYSTEM "employee.dtd">

<employee>
  <emp_id>101</emp_id>
  <emp_name>Rohan</emp_name>
  <emp_dept>IT</emp_dept>
</employee>
```

Advantages of DTD

- Simple and easy to write
- Good for small XML documents

2. XML Using XML Schema (XSD – XML Schema Definition)

What is XML Schema?

- More powerful than DTD.
- Defines:
 - Data types (int, date, string)
 - Restrictions (minLength, maxLength)
 - Namespaces
- Written using XML syntax.

Example: XML Schema + XML

employee.xsd

```
<?xml version="1.0"?>
<xs:schema xmlns:xs="http://www.w3.org/2001/XMLSchema">

  <xs:element name="employee">
    <xs:complexType>
      <xs:sequence>
        <xs:element name="emp_id" type="xs:int" />
        <xs:element name="emp_name" type="xs:string" />
        <xs:element name="emp_dept" type="xs:string" />
      </xs:sequence>
    </xs:complexType>
  </xs:element>

</xs:schema>
```

employee.xml

```
<?xml version="1.0"?>
<employee xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
  xsi:noNamespaceSchemaLocation="employee.xsd">

  <emp_id>101</emp_id>
  <emp_name>Rohan</emp_name>
  <emp_dept>IT</emp_dept>

</employee>
```

OR

Q2 a) Explain servlet session management & tracking methods. (9 marks)

Ans:

HTTP is **stateless**, meaning the server does not remember users between requests. To identify the same user across multiple requests, **session management** is required.

Explanation of Session Management in Servlets

1. What is a Session?

- A **session** is a way to store user information across multiple requests.
- It starts when a user sends the first request and ends when:
 - The browser is closed
 - Session timeout occurs
 - Logout or session.invalidate() is called

Example session data:

- Username
- Shopping cart
- Login status

? Session Tracking Methods

1. Cookies

- Small text files stored in the user's browser.
- Server sends a cookie → Browser stores it → Returns in every request.
- Helps identify the user.

Example in Servlet

```
Cookie c = new Cookie("username", "Vijay");  
response.addCookie(c);
```

To read:

```
Cookie[] ck = request.getCookies();
```

2. URL Rewriting

- Appends session ID to URL.
- Used when cookies are disabled.
- Example:

index.jsp;jsessionid=ABC123XYZ

Example in Servlet

```
String url = response.encodeURL("dashboard.jsp");
```

Advantages

- Works even when cookies are disabled

Disadvantages

- Every link must be rewritten
- Session ID visible in URL (less secure)

3. Hidden Form Fields

- A hidden field inside HTML <form> stores user data and is sent automatically.

Example:

```
<input type="hidden" name="sessionID" value="12345">
```

Disadvantages

- Works only with *form submission*
- Not secure

4. HttpSession (Most Common)

- Provided by servlet API.
- Server creates a session object and assigns a session ID.
- More secure and easy.

Creating / Reading Session

```
HttpSession session = request.getSession();  
session.setAttribute("user", "Vijay");
```

Reading:

```
String user = (String) session.getAttribute("user");
```

Q2 b) Write note on: (5 marks)

Ans:

i) AJAX (Asynchronous JavaScript and XML)

Concept in Brief

AJAX is a technique that allows web pages to **send and receive data from the server without reloading the entire page.**

Key Points

1. Stands for **Asynchronous JavaScript and XML.**
2. Uses:
 - **JavaScript** for sending requests
 - **XMLHttpRequest object** for communication
 - **JSON/XML** for data
3. Makes web pages faster and more user-friendly.
4. Example applications: Gmail auto-refresh, search suggestions, chat apps.

How AJAX Works (Steps)

1. User triggers an event (button click).
2. JavaScript creates XMLHttpRequest object.
3. Browser sends request to server in background.
4. Server sends response (JSON/XML).
5. JavaScript updates only part of the webpage.

Simple AJAX Example

```
var xhttp = new XMLHttpRequest();
xhttp.onreadystatechange = function() {
    if (this.readyState == 4 && this.status == 200) {
        document.getElementById("msg").innerHTML = this.responseText;
    }
};
xhttp.open("GET", "data.jsp", true);
xhttp.send();
```

ii) doGet() vs doPost() (Servlet)

Concept in Brief

Both methods handle HTTP requests in servlets but behave differently.

Comparison: doGet() vs doPost()

doGet()	doPost()
Used for retrieving data	Used for sending data to server
Data sent through URL	Data sent inside body
Maximum data limit (because URL size is limited)	No size limit
Less secure (URL visible)	More secure (data hidden)
Bookmarks possible	Cannot bookmark
Used for: search, reading info	Used for: login forms, file upload

Example doGet()

```
public void doGet(HttpServletRequest req, HttpServletResponse res)
{
    res.getWriter().println("This is GET");
}
```

Example doPost()

```
public void doPost(HttpServletRequest req, HttpServletResponse res)
{
    res.getWriter().println("This is POST");
}
```

Q3 a) Explain JavaBeans Classes & JSP. (8 Marks)

Ans:

JavaBeans are reusable Java classes containing properties and methods.

In **JSP**, JavaBeans are used to separate business logic from presentation logic.

JSP uses special tags (<jsp:useBean>, <jsp:getProperty>, <jsp:setProperty>) to interact with JavaBeans.

1. What are JavaBeans?

A **JavaBean** is a simple Java class used to store and manage data in a reusable form. It follows specific rules:

Features of JavaBean

1. Must have a **no-argument constructor**.
2. Class should be **public**.
3. Must use **getter and setter methods** to access private properties.
4. Should implement **Serializable** (optional).
5. Used for MVC: Model = JavaBean.

2. Example of JavaBean Class

```
public class Student implements java.io.Serializable {  
    private String name;  
    private int marks;  
  
    public Student() {} // No-arg constructor  
  
    public String getName() { return name; }  
    public void setName(String name) { this.name = name; }  
  
    public int getMarks() { return marks; }  
    public void setMarks(int marks) { this.marks = marks; }  
}
```

3. Using JavaBeans in JSP

JSP Tags for JavaBeans

Tag	Meaning
<code><jsp:useBean></code>	Create or locate bean object
<code><jsp:setProperty></code>	Set bean property
<code><jsp:getProperty></code>	Get bean property

Example (JSP page):

```
<jsp:useBean id="s" class="Student" />
```

```
<jsp:setProperty name="s" property="name" value="Vijay" />
```

```
<jsp:setProperty name="s" property="marks" value="85" />
```

Student Name: `<jsp:getProperty name="s" property="name" />``
`

Marks: `<jsp:getProperty name="s" property="marks" />`

4. Why Use JavaBeans with JSP?

1. Separates HTML (view) from Java code (model).
2. Reusable code.
3. Cleaner and maintainable JSP pages.
4. Supports MVC architecture.

Q3 b) Explain the concept of Web Services. Explain WSDL & SOAP. (9 Marks)

Ans:

Web Services allow applications to communicate over the internet using standard protocols (HTTP, XML, SOAP, WSDL).

Different systems (Java, .NET, Python) can communicate because data uses **platform-independent XML format**.

1. What are Web Services?

A **Web Service** is:

- A software component
- Accessible over the **network**
- Uses standard web protocols
- Allows **interoperability** between different applications

Characteristics

1. Platform independent (uses XML).
2. Loosely coupled communication.
3. Allows remote calls (client calls server methods).
4. Uses open standards like SOAP, WSDL.

Example uses

- Online payment APIs
- Weather API
- Login using Google/Facebook

2. SOAP (Simple Object Access Protocol)

Definition

SOAP is an XML-based protocol used to exchange messages between client and server.

Features of SOAP

1. Uses **XML** for message format
2. Works over **HTTP, SMTP**
3. Highly **secure**
4. Standardized by W3C
5. Used in enterprise-level applications

SOAP Message Structure

```
<Envelope>  
  <Header>...</Header>  
  <Body>...</Body>  
  <Fault>...</Fault>  
</Envelope>
```

Example SOAP Message

```
<Envelope>  
  <Body>  
    <getWeather>  
      <city>Pune</city>  
    </getWeather>  
  </Body>  
</Envelope>
```

3. WSDL (Web Service Description Language)

Definition

WSDL is an **XML document** that describes details about a web service — what it does, where it is located, and how to call it.

What WSDL Contains

1. **Service location (URL)**
2. **Operations (methods)**
3. **Input & output parameters**
4. **Data types**
5. **Communication protocol (SOAP)**

Example WSDL Snippet

```
<definitions>
  <message name="getWeatherRequest">
    <part name="city" type="string"/>
  </message>

  <portType name="WeatherService">
    <operation name="getWeather"/>
  </portType>
</definitions>
```

OR

Q4 a) JSP vs Servlets (Min. 04). Explain how JSPs are translated to Servlets. (8 Marks)

Ans:

☐ **Servlets** → Java programs that generate HTML using Java code.

☐ **JSP** → HTML pages with embedded Java code.

Both do the same work, but their approach is different.

1. JSP vs Servlets (Minimum 4 points)

JSP	Servlet
JSP is HTML-based with Java embedded inside.	Servlet is Java-based with HTML written inside Java code.
Best for presentation layer (UI).	Best for business logic (processing).
Easier to write and modify for designers.	Harder to modify because HTML is written inside Java print statements.
Automatically converted into a servlet internally.	Already a servlet; no conversion required.
Supports custom tags, JavaBeans, JSTL directly.	Must write code manually for everything.
Changes in JSP do not require recompilation .	Servlet changes require recompilation & redeployment.

2. How JSPs Are Translated to Servlets

JSP does not directly run.

When a JSP page is requested, the web container (Tomcat) performs **three automatic steps**:

Step 1 – Translation

- JSP file → converted to a **Java servlet file**
- Example:
index.jsp → index_jsp.java

Step 2 – Compilation

- The generated Java servlet file is compiled to a .class file.
Example:
index_jsp.java → index_jsp.class

Step 3 – Execution

- The servlet .class file is loaded.
- service(), doGet(), or doPost() is executed.
- Output is sent to browser.

JSP Lifecycle = JSP translation + servlet lifecycle

1. JSP page request
2. Translation to servlet
3. Compilation
4. Loading class
5. Instantiation
6. _jspInit()
7. _jspService()
8. _jspDestroy()

Simple Example of Generated Servlet Code

JSP:

Hello <%= name %>

Generated servlet (simplified):

```
out.write("Hello ");  
out.write(name);
```

Thus, **every JSP becomes a servlet internally.**

Q4 b) Write note on Struts. Explain Struts with example code block. (9 Marks)

Ans:

Struts is a **Java-based MVC framework** used to build web applications. It separates application into:

- **Model** (Business logic)
- **View** (JSP)
- **Controller** (ActionServlet)

It is widely used in enterprise applications.

1. What is Struts?

Struts (Apache Struts) is an open-source framework that:

- Implements **MVC** architecture
- Uses **XML configuration**
- Uses **Action classes** to process requests
- Uses **JSP** for the view
- Provides **form beans, interceptors, validators, and tag libraries**

2. Components of Struts Framework

i) **ActionServlet (Controller)**

- Main controller of the Struts application.
- Receives all user requests.

ii) **struts-config.xml**

- Configuration file that maps:
 - URLs to action classes
 - Views (JSP pages)
 - Form beans

iii) **ActionForm (Model - Data Bean)**

- Used to store input data from user.

iv) **Action Class**

- Contains business logic.
- Processes the request and returns the name of the next page.

v) **JSP (View)**

- Displays output to the user.

3. Struts Request Flow (Very Important for Exam)

1. User sends request → Controller (ActionServlet) receives it.
2. ActionServlet checks **struts-config.xml**.
3. Creates ActionForm & fills user data.
4. Calls **Action class**.
5. Action class returns **ActionForward** (next page).
6. JSP page is displayed to user.

4. Struts Example Code Block

(i) struts-config.xml

```
<action path="/login"
        type="com.app.LoginAction"
        name="loginForm"
        input="/login.jsp"
        scope="request">
    <forward name="success" path="/welcome.jsp"/>
    <forward name="fail" path="/error.jsp"/>
</action>
```

(ii) ActionForm (LoginForm.java)

```
public class LoginForm extends ActionForm {
    private String username;
    private String password;

    // getters and setters
}
```

(iii) Action Class (LoginAction.java)

```
public class LoginAction extends Action {
    public ActionForward execute(ActionMapping mapping,
                                ActionForm form,
                                HttpServletRequest request,
                                HttpServletResponse response) {

        LoginForm lf = (LoginForm) form;

        if(lf.getUsername().equals("admin") && lf.getPassword().equals("123")) {
            return mapping.findForward("success");
        } else {
            return mapping.findForward("fail");
        }
    }
}
```

(iv) View (welcome.jsp)

```
<html><body>
<h2>Login Successful!</h2>
</body></html>
```

Q5) a) Explain MySQL & PHP for inserting student data into the Students table. [8]

Ans:

PHP can insert data into a MySQL database using **MySQLi** or **PDO**. Steps include: connect → write INSERT query → execute → confirm success.

1) Steps to Insert Data Using PHP + MySQL

Step 1: Create Students Table

(Assume suitable fields)

```
CREATE TABLE students (  
    roll_no INT PRIMARY KEY,  
    name VARCHAR(50),  
    dept VARCHAR(30),  
    marks INT  
);
```

Step 2: HTML Form to Accept Student Data

```
<form action="insert.php" method="post">  
    Roll No: <input type="text" name="roll"><br>  
    Name: <input type="text" name="name"><br>  
    Department: <input type="text" name="dept"><br>  
    Marks: <input type="text" name="marks"><br>  
    <input type="submit" value="Insert">  
</form>
```

Step 3: PHP Code to Insert Data (insert.php)

```
<?php
// Step 1: Collect Data
$roll = $_POST['roll'];
$name = $_POST['name'];
$dept = $_POST['dept'];
$marks = $_POST['marks'];

// Step 2: Connect to MySQL
$conn = mysqli_connect("localhost", "root", "", "college");

// Step 3: Check connection
if(!$conn){
    die("Connection Failed: " . mysqli_connect_error());
}

// Step 4: Write Insert Query
$sql = "INSERT INTO students (roll_no, name, dept, marks)
        VALUES ('$roll', '$name', '$dept', '$marks')";

// Step 5: Execute Query
if(mysqli_query($conn, $sql)){
    echo "Student Record Inserted Successfully!";
} else {
    echo "Error: " . mysqli_error($conn);
}

// Step 6: Close connection
mysqli_close($conn);
?>
```

Explanation

1. PHP sends form data to server using POST.
2. MySQLi functions connect to database.
3. INSERT query is prepared with form values.
4. Query is executed using `mysqli_query()`.
5. On success, a message is displayed.
6. Connection is closed using `mysqli_close()`.

Q5) b) Write note on: ASP.NET & NodeJS [5 + 5]

Ans:

i) ASP.NET – Short Note

Concept in Brief

ASP.NET is a **server-side web development framework** by Microsoft.

Explanation

- Part of the **.NET Framework**.
- Used to create **dynamic websites, web apps, and web services**.
- Supports languages like **C#, VB.NET**.
- Provides rich **server controls** (GridView, TextBox, Button).
- Follows **code-behind architecture** (HTML and C# separate).
- Supports **session handling, caching, security, and state management**.
- Works with SQL Server, MySQL, Oracle databases.
- Provides **ASP.NET MVC, Web Forms, Web API, Razor Pages**.

ii) NodeJS – Short Note

Concept in Brief

NodeJS is a **JavaScript runtime** that lets JS run on the **server side**.

Explanation

- Uses **Google V8 JavaScript Engine**.
- Based on **event-driven, non-blocking I/O**.
- Very fast and scalable.
- Ideal for **real-time applications** (chat, notifications, streaming apps).
- Uses **npm** (Node Package Manager) with thousands of libraries.
- Popular frameworks: **ExpressJS, NestJS**.
- Supports asynchronous operations using **promises, async/await**.

Q6) a) Explain PHP functions and arrays in PHP. [10]

Ans:

1) PHP Functions

Concept

Functions are reusable blocks of code that perform specific tasks.

Features

- Avoid repetition
- Improve readability
- Can return values
- Accept parameters

Example

```
<?php
function addNumbers($a, $b){
    return $a + $b;
}
echo addNumbers(10, 20);
?>
```

2) PHP Arrays

Types of Arrays

1. **Indexed Array**
2. **Associative Array**
3. **Multidimensional Array**

Examples

Indexed Array

```
$colors = array("Red", "Blue", "Green");
```

Associative Array

```
$student = array("name"=>"Amit", "age"=>21);
```

Multidimensional Array

```
$marks = array(  
    array(101, "Amit", 89),  
    array(102, "Rohit", 76)  
);
```

Q6) b) Explain the concept of .NET Framework and C#. [8]

Ans:

1) .NET Framework

Concept

A Microsoft platform used to build web, desktop, and enterprise applications.

Main Components

1. CLR (Common Language Runtime)

- Executes .NET programs
- Manages memory
- Garbage collection
- Thread management

2. BCL (Base Class Library)

- Predefined classes (Collections, File I/O, XML, Networking)

3. ASP.NET

- Web development framework

4. Multiple Language Support

- C#
- VB.NET
- F#

2) C# (C-Sharp)

Concept

C# is an **object-oriented programming language** used in .NET.

Features

- Strongly typed
- Supports OOP concepts (class, object, inheritance, polymorphism)
- Automatic memory management
- Exception handling
- Modern syntax similar to Java

Example

using System;

```
class Demo {  
    static void Main() {  
        Console.WriteLine("Hello C#");  
    }  
}
```

Q7) a) Explain Ruby with respect to scalar types & their operations, control statements & arrays.[12]

Ans:

1) Scalar Types in Ruby

Scalar types are the **basic data types** which store single values.

i) Integer

- Used for whole numbers.

Example:

```
a = 10
```

```
b = 5
```

```
sum = a + b # 15
```

ii) Float

- Used for decimal numbers.

```
x = 5.5
```

```
y = 2.2
```

```
z = x * y
```

iii) String

- Sequence of characters inside quotes.

```
name = "Vijay"
```

```
msg = "Hello " + name
```

iv) Boolean

- Stores **true** or **false**.

```
is_valid = true
```

v) Nil

- Represents no value.

```
data = nil
```

Operations on Scalar Types

- **Arithmetic:** +, -, *, /, %
- **String operations:** concatenation (+), repetition (*), interpolation
- "Hello #{name}"
- **Comparison:** ==, !=, >, <, >=, <=
- **Logical:** &&, ||, !

2) Control Statements in Ruby

i) If – Else

```
age = 18
if age >= 18
  puts "Eligible"
else
  puts "Not eligible"
end
```

ii) Unless (opposite of if)

```
unless age < 18
  puts "Adult"
end
```

iii) Case Statement

Used like switch-case in other languages.

```
day = 2

case day
when 1
  puts "Monday"
when 2
  puts "Tuesday"
else
  puts "Other day"
end
```

iv) Loops

For Loop

```
for i in 1..5
  puts i
end
```

While Loop

```
i = 1
while i <= 5
  puts i
  i += 1
end
```

Times Loop

```
5.times do
  puts "Hello"
end
```

3) Arrays in Ruby

Array is an **ordered list** that can store multiple values (even different types).

Creating array

```
arr = [10, "Ruby", 3.14]
```

Accessing elements

```
puts arr[0]  # 10  
puts arr[1]  # Ruby
```

Array operations

- **Add element**
- `arr << "New"`
- **Delete element**
- `arr.delete(10)`
- **Iterating array**
- `arr.each do |item|`
- `puts item`
- `end`

Array methods

- `length`
- `push`
- `pop`
- `sort`
- `reverse`

Q7 b) Write note on EJB (Enterprise Java Beans).[5]

Ans:

EJB (Enterprise Java Bean) is a **server-side component** used to build **large, distributed, scalable enterprise applications** in Java.

Features of EJB

- **Reusable components**
- **Distributed computing**
- **Automatic transaction management**
- **Security management**
- **Remote method invocation**

Types of EJB

1. Session Beans

- Perform business logic.
- Types:
 - *Stateless* (no data stored between calls)
 - *Stateful* (stores client data)
 - *Singleton* (one instance for whole application)

2. Entity Beans

- Represent database tables (older versions, replaced by JPA).

3. Message Driven Beans

- Used for asynchronous communication (messages).

Advantages

- Reduces programming complexity
- Provides security, transactions automatically
- Good for banking, e-commerce, large systems

OR-Q8) a) Explain Rails Applications & databases. Also explain Rails with AJAX.[12]

Ans:

1) Rails Applications & Databases

Ruby on Rails follows **MVC architecture**:

i) Model

- Handles **database interaction**.
- Uses **ActiveRecord**, which automatically maps tables to classes.

Example:

```
class Student < ApplicationRecord
end
```

This automatically maps to students table.

ii) View

- Displays output to user (HTML, CSS, ERB templates).

iii) Controller

- Receives request, processes data, calls model & view.

```
class StudentsController < ApplicationController
  def index
    @students = Student.all
  end
end
```

Connecting Rails Application to Database

Database Configuration

In config/database.yml:

development:

adapter: mysql2

database: students_db

username: root

password: ""

CRUD operations using ActiveRecord

- **Create**
- Student.create(name: "Vijay", age: 20)
- **Read**
- Student.all
- **Update**
- s = Student.find(1)
- s.update(age: 22)
- **Delete**
- Student.find(1).destroy

ActiveRecord makes database operations very easy.

2) Rails with AJAX

AJAX allows **partial page updates** without reloading the page.

How Rails supports AJAX

Rails provides:

- remote: true option for forms & links
- .js.erb files for dynamic JavaScript responses

Example: AJAX Form

Form:

```
<%= form_with url: "/students", remote: true do |f| %>
  <%= f.text_field :name %>
  <%= f.submit "Save" %>
<% end %>
```

Controller:

```
def create
  @student = Student.new(student_params)
  @student.save
end
```

Response file: create.js.erb

```
alert("Student saved successfully!");
```

Advantages of Rails AJAX

- Faster updates
- No full page reload
- Better user experience

Q8 b) Explain advantages of Ruby & Rails.[5]

Ans:

Advantages of Ruby

- Very **simple and readable syntax**
- Pure **object-oriented language**
- Dynamic and flexible
- Large number of built-in methods

Advantages of Rails

- Follows **MVC architecture**
- **Rapid application development**
- Built-in support for **database handling** via ActiveRecord
- **Convention over configuration** reduces developer effort
- **Security features** built-in
- Strong community support

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Best Regards, **Vishwajeet Londhe**